Lecture 32

STAT 109: Introductory Biostatistics — One-way ANOVA

Lecture 32

One-way ANOVA — comparing several means

When we have one numerical response and one categorical explanatory variable with more than two levels (for example three randomly assigned treatments), a one-way analysis of variance (ANOVA) tests the omnibus null hypothesis that the population mean of the response is the same in every group:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

The alternative is that **at least one** group mean differs from the others (without saying which pair).

In R, one-way ANOVA for balanced or unbalanced groups is often fit as a linear model with only the grouping factor, then summarized with `anova()`:

- `fit <- lm(response ~ group, data = ...)`
- `anova(fit)` prints an **F-test** of whether `group` explains a detectable amount of variation in `response`.

Conditions. As in the two-sample t-setting, we need **independent** observations (often supported by **random assignment** in an experiment). Populations should be **roughly normal** within each group, and **similar spread** across groups; very unequal spreads or strong skew or small sample sizes can make the F-test untrustworthy.

Connection to Lecture 25. If there are only **two** groups, the one-way ANOVA F-test is equivalent (for equal variance assumptions) to a pooled two-sample t-test; with **Welch** unequal variances we use `t.test(..., var.equal = FALSE)` instead.

Example 1 — RCT: three soil amendments and stem height (cm)

Story. In a greenhouse trial, seedlings were randomly assigned to one of three soil-amendment mixes (A, B, C). After six weeks, stem height (cm) was measured once per plant.

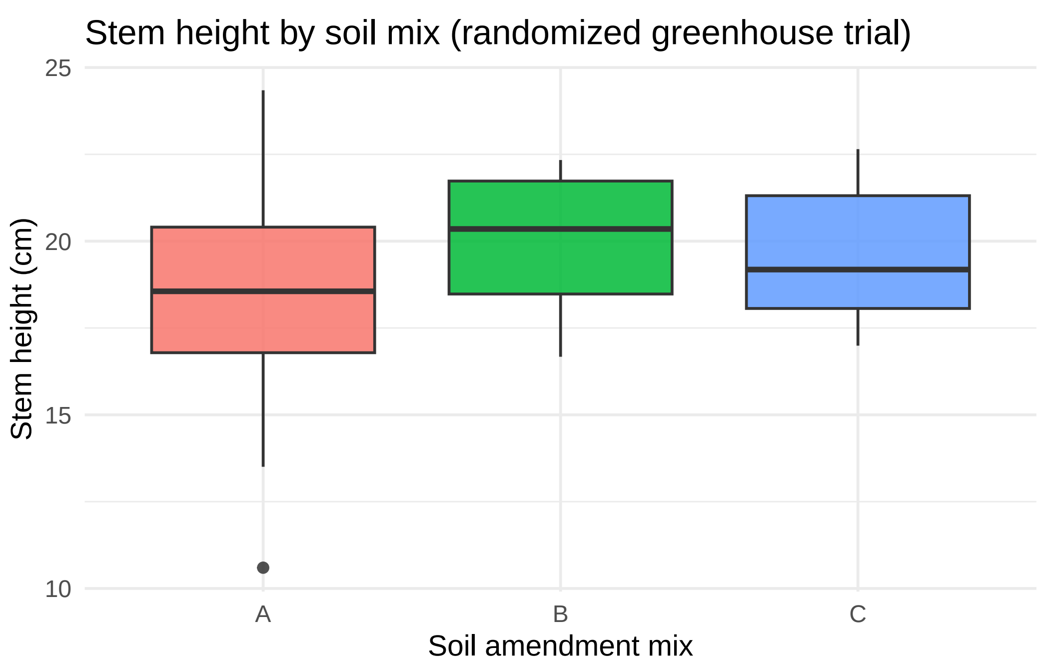
```r
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

set.seed(3201)
n_per <- 22
mu <- c(A = 18, B = 21, C = 19.5)
sd_m <- c(A = 2.4, B = 2.2, C = 2.3)
soil_rct <- tibble(
  mix = factor(
    sample(rep(c("A", "B", "C"), each = n_per)),
    levels = c("A", "B", "C")
  )
) |>
  mutate(height_cm = rnorm(n(), mean = mu[mix], sd = sd_m[mix]))
```

```
soil_rct |>
  group_by(mix) |>
  summarise(n = n(), mean_cm = mean(height_cm), sd_cm = sd(height_cm), .groups = "drop")
```

```
# A tibble: 3 x 4
  mix       n mean_cm sd_cm
  <fct> <int>   <dbl> <dbl>
1 A        22    18.1  3.09
2 B        22    20.1  1.85
3 C        22    19.5  1.75
```

Boxplot by randomly assigned mix

```
ggplot(soil_rct, aes(x = mix, y = height_cm, fill = mix)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.85) +
  labs(
    title = "Stem height by soil mix (randomized greenhouse trial)",
    x = "Soil amendment mix",
    y = "Stem height (cm)"
  ) +
  theme_minimal()
```



One-way ANOVA: `lm` + `anova`

```
fit_soil <- lm(height_cm ~ mix, data = soil_rct)
anova(fit_soil)
```

```
Analysis of Variance Table

Response: height_cm
          Df Sum Sq Mean Sq F value  Pr(>F)
```

```
mix        2  44.33 22.1669  4.1508 0.02026 *
Residuals 63 336.44  5.3404
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Read the output. The row `mix` tests whether stem height differs across mixes on average, assuming the ANOVA conditions are met. A small *p*-value suggests the data are not compatible with equal population means across all three mixes under the test's assumptions.

Example 2 — RCT: three stretching routines and flexibility score

Story. Office workers volunteered for a wellness study; each was randomly assigned to one of three five-minute stretching routines (R1, R2, R3). A flexibility score (0–50) was recorded after two weeks.

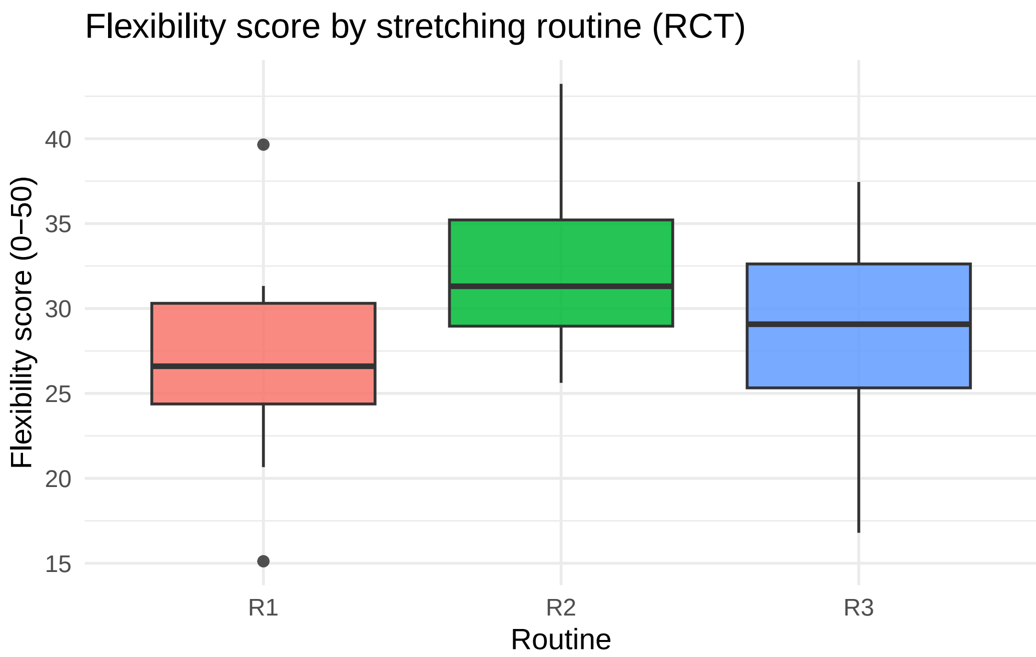
Research Question. Does the mean score differ between at least two of the routines for people like those who enrolled?

```r
set.seed(3202)
n_per <- 18
mu_s <- c(R1 = 28, R2 = 32, R3 = 30)
sd_s <- c(R1 = 5, R2 = 4.5, R3 = 4.8)
stretch_rct <- tibble(
  routine = factor(
    sample(rep(c("R1", "R2", "R3"), each = n_per)),
    levels = c("R1", "R2", "R3")
  )
) |>
  mutate(score = pmin(pmax(rnorm(n(), mean = mu_s[routine], sd = sd_s[routine]), 0), 50))

stretch_rct |>
  group_by(routine) |>
  summarise(n = n(), mean_score = mean(score), sd_score = sd(score), .groups = "drop")
```

```
# A tibble: 3 x 4
  routine     n mean_score sd_score
  <fct>   <int>      <dbl>    <dbl>
1 R1         18       27.0     5.27
2 R2         18       32.6     5.03
3 R3         18       28.6     5.37
```

Boxplot by randomly assigned routine

```r
ggplot(stretch_rct, aes(x = routine, y = score, fill = routine)) +
  geom_boxplot(show.legend = FALSE, alpha = 0.85) +
  labs(
    title = "Flexibility score by stretching routine (RCT)",
    x = "Routine",
    y = "Flexibility score (0-50)"
  ) +
  theme_minimal()
```



One-way ANOVA: `lm` + `anova`

```
fit_stretch <- lm(score ~ routine, data = stretch_rct)
anova(fit_stretch)
```

```
Analysis of Variance Table

Response: score
          Df Sum Sq Mean Sq F value   Pr(>F)
routine    2  299.7 149.852  5.4846 0.006959 **
Residuals 51 1393.4  27.322
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion: